

# FILTER DESIGN TEST CASE

Lattice Filter - Band 12 (700 MHz) for 4G LTE Carrier Aggregation  
Design Validation and Performance Characterization

## DOCUMENT INFORMATION

|                  |                    |
|------------------|--------------------|
| Document Number: | FD-TC-0022         |
| Revision:        | Rev A.2            |
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| Reviewer:        | Dr. Stefan Richter |
| Design Center:   | Singapore RF Lab   |
| Created Date:    | 2024-05-29         |
| Last Modified:   | 2024-06-05         |
| Status:          | CONDITIONAL PASS   |
| Priority:        | P2 - High          |

## 1. OBJECTIVE

This document describes the design, simulation, and characterization of a Lattice Filter filter targeting Band 12 (700 MHz) for 4G LTE Carrier Aggregation. The filter is designed on AlN on Si substrate with a target center frequency of 2366.6 MHz and bandwidth of 50.8 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 4G LTE Carrier Aggregation module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

## 2. SCOPE

- Filter Type: Lattice Filter
- Frequency Band: Band 12 (700 MHz)
- Application: 4G LTE Carrier Aggregation
- Substrate: AlN on Si
- Package: SiP (System in Package)
- Design Tool: Sonnet EM Suite v16
- Design Center: Singapore RF Lab

This specification applies to all variants of the Lattice Filter design targeting Band 12 (700 MHz). Results are used to qualify the design for mass production release.

## 3. DESIGN PARAMETERS

|                          |                |
|--------------------------|----------------|
| Center Frequency:        | 2366.6 MHz     |
| Bandwidth (3 dB):        | 50.8 MHz       |
| Insertion Loss Target:   | $\leq 3.2$ dB  |
| Return Loss Target:      | $\geq 20.0$ dB |
| Out-of-Band Rejection:   | $\geq 49.0$ dB |
| Isolation (Duplexer):    | $\geq 36.0$ dB |
| Q Factor:                | 6627           |
| Temperature Coefficient: | -8.3 ppm/°C    |
| Die Size:                | 1.9 x 1.6 mm   |
| Wafer Size:              | 6-inch         |
| Number of Resonators:    | 6              |
| Electrode Material:      | Al             |

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Sonnet EM Suite v16 and load the Lattice Filter template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 2366.6 MHz, BW 50.8 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, AlN on Si).
10. Perform on-wafer probing using Keysight N5227B PNA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 7100 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.2 dB
- Return loss in passband shall be >= 20.0 dB
- Out-of-band rejection at +/- 76 MHz offset >= 49.0 dB
- Group delay variation across passband shall be <= 2 ns
- Temperature drift over -40°C to +85°C shall be <= 2.5 MHz
- Power handling shall be >= +35 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 90%

6. TEST RESULTS

Measurement Date: 2024-06-05  
Devices Tested: 57  
Insertion Loss (meas): 2.93 dB  
Return Loss (meas): 21.8 dB  
Rejection (meas): 52.0 dB  
Isolation (meas): 41.2 dB  
Q Factor (meas): 6627  
Group Delay Variation: 5.3 ns  
Power Handling: +32 dBm  
Die Yield: 80.4%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The Lattice Filter design demonstrated acceptable performance for Band 12 (700 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Package parasitic effects were consistent with 3D EM model predictions.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

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- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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|-----------|--------------------|-------|------------|
| Author:   | Mark Sullivan      | Date: | 2024-06-05 |
| Reviewer: | Dr. Stefan Richter | Date: | 2024-06-11 |
| Approver: | Dr. Hans Weber     | Date: | 2024-06-12 |

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